PROJECT MANAGEMENT & SCHEDULE SPECIFICATION

HELPBOT: ENGINEERING PROJECT PLAN

Work Breakdown Structure, Critical Path Method (CPM), Gantt Baseline & Resource Loading

1. Document Control & Scheduling Baseline

Project Reference	Case Study No. 97 — Regional Broadband Service Provider	**Version Baseline**	1.0 (Schedule Baseline)
Total Capital Outlay	₹18,00,000 (₹18 Lakh Fixed Ceiling)	**Target Schedule Window**	14 Calendar Weeks (70 Working Days on 5d/wk)
Core Engineering Staff	1 Project Manager / Lead, 2 NLP & Integration Engineers, 1 Full-Stack QA Engineer		

2. Work Breakdown Structure (WBS)

WBS Code	Work Package Title	Engineering Deliverable & Scope Boundary
1.0	HelpBot Implementation	End-to-end multi-channel conversational platform across Web & WhatsApp.
1.1	Requirements & Architecture	IEEE 830 SRS, boundary rules, MoSCoW prioritization, and UML models.
1.2	NLP Engine & Elicitation	Domain dataset curation, intent classifiers (5 drivers), and confidence scoring.
1.3	Backend Adapter Engineering	Loose REST adapter for Billing API, OLT network diagnostics, and NOC outage bus.
1.4	Handover & Channel Ingress	WhatsApp Cloud API webhook, Web widget, and agent CRM routing engine.
1.5	Verification & Quality Gates	Boundary Value Analysis (BVA), decision tables, load testing, and DRE profiling.
1.6	Pilot Cohort & Cutover	Call-centre agent enablement, shadow testing, cohort cutover, and warranty buffer.

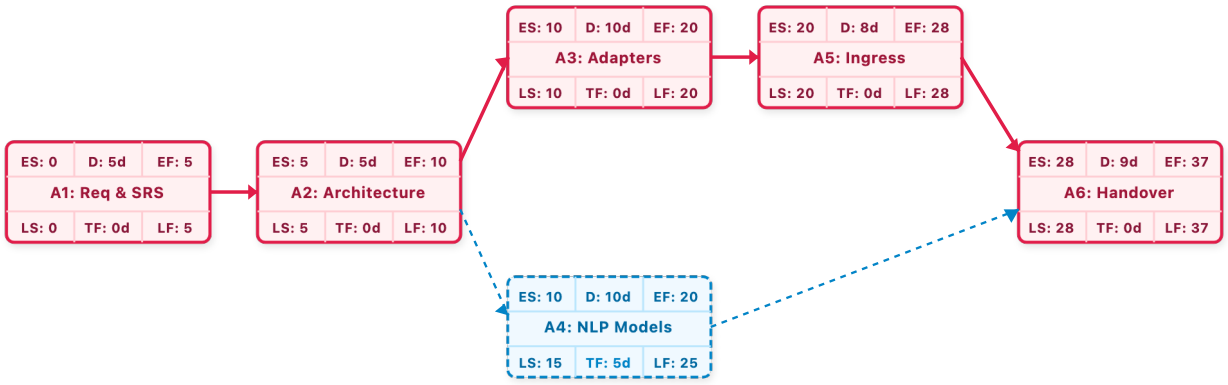
3. Activity Table & Critical Path Calculations

Early Start (ES), Early Finish (EF), Late Start (LS), Late Finish (LF), and Total Float (TF) are computed across a 5-day work week. Critical activities exhibit **Total Float = 0**.

ID	Activity Description	Effort	Dur	Pred	ES	EF	LS	LF	Float	Status
A1	Requirements & SRS Baseline	15 p-d	5 d	None	0	5	0	5	0 d	CRITICAL
A2	Architecture & UML Design	15 p-d	5 d	A1	5	10	5	10	0 d	CRITICAL
A3	Billing & Outage REST Adapters	30 p-d	10 d	A2	10	20	10	20	0 d	CRITICAL
A4	NLP Intent & Confidence Model	25 p-d	10 d	A2	10	20	15	25	5 d	FLOAT: 5D
A5	Multi-Channel Ingress Webhooks	16 p-d	8 d	A3	20	28	20	28	0 d	CRITICAL
A6	Agent Handover & Context Desk	18 p-d	9 d	A4, A5	28	37	28	37	0 d	CRITICAL
A7	Integration & BVA Test Suite	20 p-d	10 d	A6	37	47	37	47	0 d	CRITICAL
A8	Agent Training & Change Mgmt	8 p-d	8 d	A6	37	45	42	50	5 d	FLOAT: 5D
A9	Pilot Deployment (10% Traffic)	12 p-d	6 d	A7, A8	47	53	47	53	0 d	CRITICAL
A10	Full Production Cutover	10 p-d	5 d	A9	53	58	53	58	0 d	CRITICAL

4. Activity-on-Node (AON) Critical Path Network Diagram

Red boxes denote critical activities where Total Float = 0. Blue dashed nodes denote non-critical paths with schedule float.



Follow-on Critical Path: A6 → A7 (QA/BVA) → A9 (Pilot) → A10 (Cutover) → A11 (Warranty Buffer = Day 63)

5. 14-Week Gantt Schedule & Key Governance Milestones



6. Milestone Quality Gates & Resource Allocation Matrix

Milestone	Title & Target Timeline	Mandatory Exit Gate Deliverable
M1	SRS & Boundary Approval (End W1)	100% sign-off from CEO & Call-Centre Manager on the 5-domain boundary.
M2	Adapter Integration Gate (End W4)	Successful tokenized balance retrieval from Billing REST API under ≤ 1.8s.
M3	Verification Sign-off (End W9)	Zero Critical/High defects; DRE ≥ 80% achieved; Handover SLA ≤ 3.0s validated.
M4	Go-Live Production Cutover (Mid W12)	Multi-channel production launch across Web & WhatsApp; 5-day warranty buffer active.

Resource Profile	Primary Ownership Area	Allocated Capacity	Workload Balance
Project Manager / Lead	Stakeholder governance, risk register, daily standups, SLA compliance	Weeks 1–14 (100%)	Balanced
Senior NLP Engineer	Intent classification, confidence scoring (C ≥ 0.70 threshold), dialogue models	Weeks 1–11 (100%)	Balanced
Backend Integrator	Billing adapter, OLT speed probe client, WhatsApp webhook, handover router	Weeks 2–12 (100%)	Balanced
Full-Stack QA Engineer	BVA test execution, handover decision tables, JMeter load tests, defect logging	Weeks 5–14 (100%)	Balanced